

ENG 101

LESSON 20

Steps in Problem Solving

Can a computer solve problems? Definitely not. It is a machine that carries out the procedures which the programmer gives it. It is the programmer then who solves the problems. There are a few steps that one has to follow in problem solving:

Step 1. The programmer must define the problem clearly. This means that he or she has to determine, in a general way, how to solve the problem. Some problems are easy, while others take months of study. The programmer should always start by asking: 'Do I understand the problem'.

Step 2. The programmer must formulate an algorithm, which is a straightforward sequence of steps of instructions used to solve the problem. Constructing an algorithm is the most important part of problem solving and is usually time-consuming. An algorithm can be described by a flowchart, which may be stated in terms of a sequence of precise sentences, or a block diagram. The latter is a diagrammatic representation of the sequence of events to be followed in solving the problem. The relationship between the events is shown by means of a connecting arrow. A block diagram can show if a process has to be repeated or if there are alternative routes to be taken.

Step 3. The programmer must translate the algorithm or flowchart into a computer program. To do so, he or she writes detailed instructions for the computer, using one of the many computer languages available following the exact sequence of the flowchart algorithm. The program is usually written on coding sheets which have a specific format drawn on them.

Step 4. The programmer must then keypunch the program, or give the coding sheets to the keypunch Operator to do it. The program is either punched on cards or entered into the computer at a terminal with a visual display unit.

Step 5. The program must then be tested. To do so, the computer operator puts the deck of cards in the card reader and presses the 'read' button. This transfers the information to the memory of the computer. Next, a printout shows if the program works or if it has errors (called bugs). If the programmer is using a terminal instead of cards to enter the instructions it is possible, with the aid of a few commands, to store the program in the memory of the computer and get a printout.

Step 6. The last step is to add the data to the program and run the job completely. The computer will then perform the calculations necessary to solve the problem. It will follow the instructions in the program to the minutest details. Therefore, one can say that the computer is a robot. It doesn't think, but simply does what it is told.

Exercises

1. Main idea

Which statement best expresses the main idea of the text? Why did you eliminate the other choices?

- ☐ 1. Constructing an algorithm is the basic step in solving a problem.
- ☐ 2. Solving problems becomes easier if certain steps are followed.
- ☐ 3. The computer does what the programmer tells it to do.

2. Understanding the passage

Decide whether the following statements are true or false (T/F) by referring to the information in the text. Then make the necessary changes so that the false statements become true.

1. The computer is a great help to people because it solves there problems.
2. All problems are equally difficult to solve.
3. An algorithm is a sequences of instructions used to solve a problem.
4. The most important part of problem-solving is defining the problem clearly.
5. Block diagrams cannot show relationships.
6. Coding sheets are used for writing programs.
7. Punched cards are the only way of transferring the program to the computer memory.
8. If the data is not added to the program, the computer cannot perform calculations.
9. It is a good idea to test the program before adding the data.
10. A computer is very intelligent. It is capable of thinking.

3. Locating Information

Find the passages in the text where the following ideas are expressed. Give the paragraph references.

- 1. Programs are usually written on certain lined forms.
-2. A block diagram can show a decisions with two different outcomes.
-3. The programmer is the one who solves the problems.
-4. Even if the programmer is using a terminal instead of cards, it is possible to get permanent copy of his program.
-5. Not all problems are of the same level of difficulty.

4. Contextual Reference

Look back at the text and find out what the words in bold typeface refer to.

1. **It** is a machine (l.1).....
2. **which** the programmer give it (l.2).....
3. **who** solves the problems (l.3).....
4. **which** may be stated (l.14).....
5. The **latter** is a diagrammatic representation (l.15).....
6. operate to do **it** (l.27).....
7. **This** transfers the information (l.32).....
8. or if **it** has errors (l.33).....
9. **It** will follow the instructions (l.40).....
10. does what **it** is told (l.42).....

5. Understanding Words

Refer back to the text and find synonyms for the following words.

1. construct (l.10).....
2. takes a lot of time (l.13).....
3. exact (1.15).....
4. mistakes (l.34)
5. help (l.35).....

Now refer back to the text and find antonyms for the following words.

6. ambiguously (l.5).....
7. specific (l.6).....
8. partially (l.39).....

6. Word forms

First choose the appropriate form of the words to complete the sentences. Then check the differences of meaning in your dictionary.

Procedure, proceed, proceeding

- a) The machine carries out the..... which the programmer gives it.

- b) You should with care when using a calculator.
2. program, programmer, programmed, programming
- a) I would like toin COBOL.
- b) There were quite a few errors in my.....
- c) My calculator is it plays a tune on the hour.
- d) Fortran is one of the many languages available on the market.
- e) Computer..... is a new field of study at the university.
- f) He is a good because he always constructs algorithm for his problems.
3. relationships, relate, related
- a. The first two steps in your program are not they are basically different.
- b. In a flowchart the.....between events is shown by means of connecting arrows.
- c. Assembler is one example of a machine.....
4. code, coding
- a. Do you have any sheet left.
- b. I have to my program.
- c. Assembler is one example of a machine.....
5. printer, printing, print, printed
- a. your name and address in block letters.
- b. was introduced by Gutenberg in Germany.
- c. The quality of the output from a daisy wheel printer is superior to that from a dot matrix.
- d.provided a hard copy of the results of data processing operations.
1. Special forms which are usually used for writing programs are called
2. Another word for program errors is.....
3. A number of steps used in solving a program is called an
4. A machine which is incapable of thinking but follows instructions is called a
5. A is either a group of exact sentences to solve a problem or a block diagram.

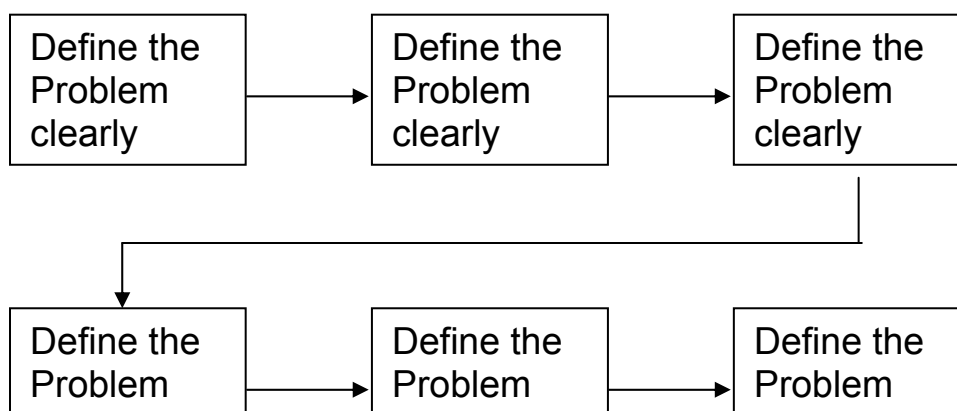
7a. Content Review

Try to think of a definition for each of these items before checking them in the Glossary. Then complete the following statements with the appropriate words. Make sure you use the correct form, i.e. singular or plural.

algorithm	coding sheets	flowchart
robot	printout	bugs

7b. Control review

Solving a problem is a process involving various steps. Complete the following diagram to show the sequence of these steps.



8.Focus Review

Focus On: Cause and effect/result

The following sentences were taken from the text on 'Steps in Problem Solving'. While reading these sentences, underline the cause once, the effect/result twice, then circle the causal-effect/result marker.

1. If the programmer is using a terminal instead of cards to enter his instructions, he can, with the aid of a few commands, store the program in the memory of the computer and get a printout.

2. The computer will follow the instructions in the program to the minutest details. Therefore, one can say that the computer is a robot.

A computer has four basic components: input, processor, memory, and output. The CPU consists of two parts: the which directs and controls the signals and commands inside the processor, and the unit which does the arithmetic operations and the decision-making operations. While the

.....is made up of a, a, a..... and a....., the is composed of, a..... and, which compares information and makes decision based on the results of the comparisons.

In a computer, internal memory or refers to the storage location inside the computer, whereas..... refers to the storage embodied in the peripherals..... can be divided into three types:.....,

..... and on the other hand may be grouped as (.....) or (.....).

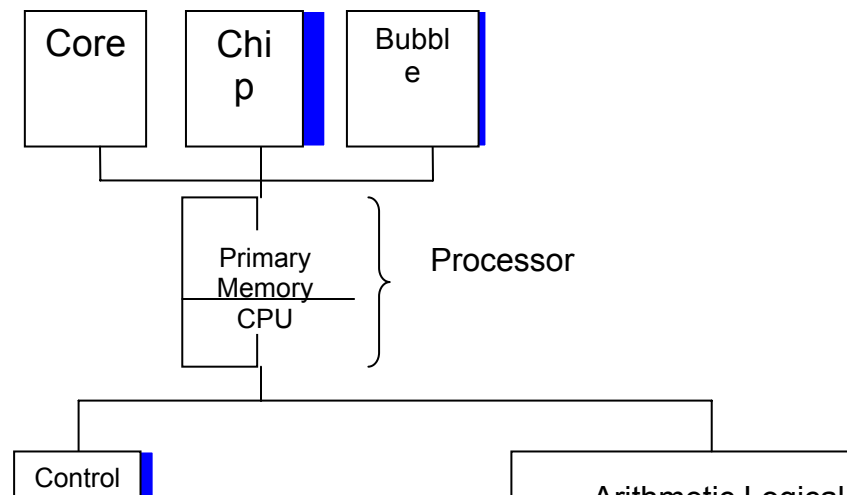
The devices can be either a, a, a..... or a These devices enter the information into the computer. After the processor has operated on it, the devices display the results of the computations on either a or a, or store them on tape or disk for future use.

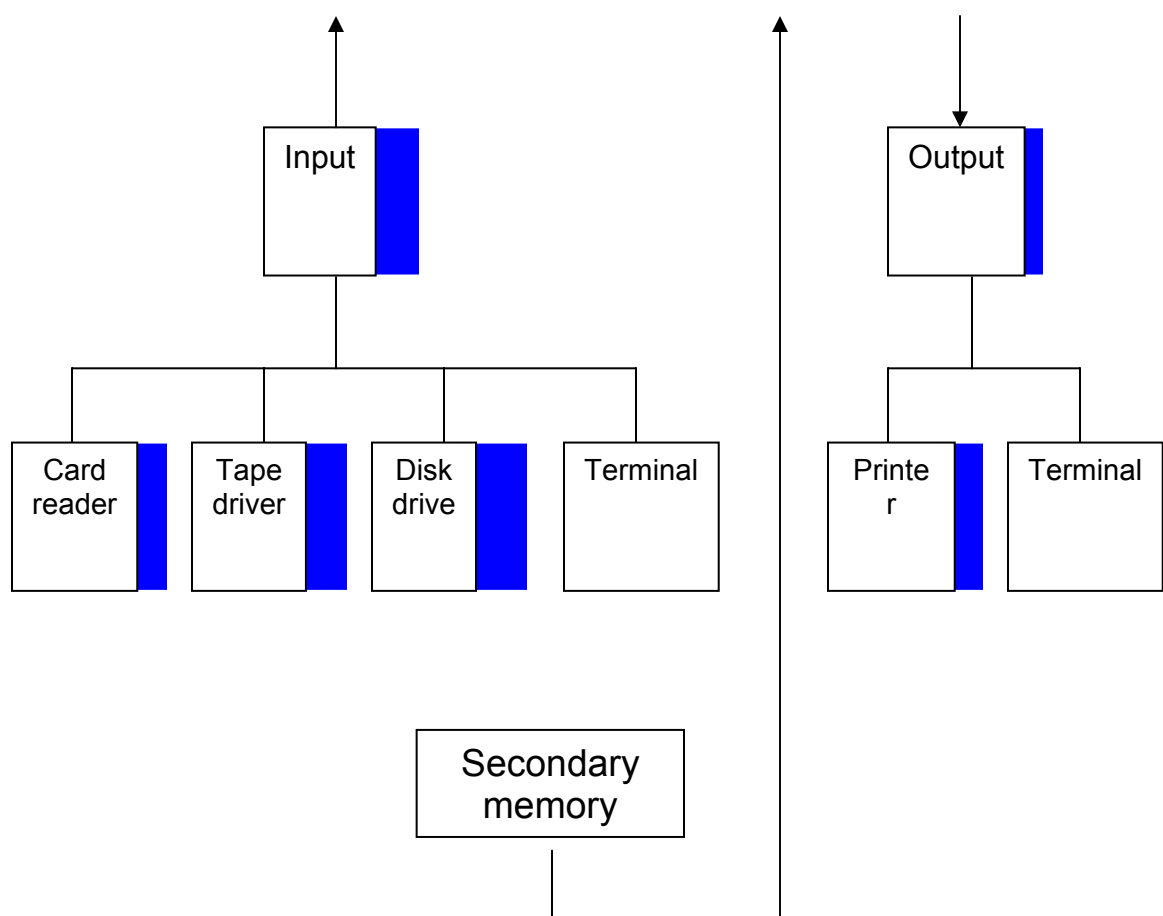
Exercise 2

Using the diagram below, complete the paragraph that follows.

A computer system

CPU





A computer has four basic Components: input, processor, memory, and output. The CPU consists of two parts: the _____ which directs and controls the signals and commands inside the processor, and the _____ unit which does the arithmetic operations and the decision making operations. While the _____ is made up of a _____ a _____, a _____, and a _____, the _____ is composed of _____, a _____ and _____, which compares information and makes decisions based on the results of the comparisons.

In a Computer internal memory or _____ refers to the storage locations inside the computer, whereas _____ refers to the storage embodied in the peripherals _____ can be divided into three types: _____, _____ and _____.

_____ on the other hand may be grouped as _____ (_____) or _____ (_____).

The _____ devices can be either a _____, a _____, a _____ or a _____.

These Devices enter in information into the computer. After the processor has operated on it, the _____ devices display the results of the computations on either a _____ or a _____, or store them on tape or disk for future use.

